

A Reference Architecture for Improving Security and Privacy in Internet of Things Applications

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Abstract – *As the promise of the Internet of Things (IoT) materializes in our everyday lives, we are often challenged with a number of concerns regarding the efficacy of the current data privacy solutions that support the pervasive components at play in IoT. The privacy and security concerns surrounding IoT often manifests themselves as a treat to end-user adoption and negatively impacts trust among end-users in these solutions. In this paper, we present a reference software architecture for building cloud-enabled IoT applications in support of collaborative pervasive systems aimed at achieving trustworthiness among end-users in IoT scenarios. We present a case study that leverages this reference architecture to protect sensitive user data in an IoT application implementation and evaluate the response of an end-user study accomplished through a survey.*

Index Terms – Software Reference Architecture, Cloud-Enabled Service Privacy and Security, Internet of Things, Collective Intelligence

I. INTRODUCTION

Trends pertaining to the use of the *Internet of Things* (IoT) to collaboratively solve complex problems in healthcare monitoring, online web site advertising, and smart home implementation scenarios, among others, are expected to continue to grow rapidly. Nonetheless, one of the most critical deterrents to mainstream user adoption of IoT systems is a looming distaste for how data privacy is enforced among the various collaborative systems at play in IoT applications. Beyond the privacy and security concerns encircling IoT systems, it is becoming more and more inevitable for pervasive collaborative devices to leverage web services for data sharing and communication to backend storage systems. With the advent of cloud computing, it is not uncommon for the mobile services that these pervasive devices communicate with, to be hosted in the *cloud*. Consequently, with the imminent domain-specific privacy and security concerns for cloud computing, IoT, pervasive systems and web services, it is important to establish a reference architecture that provides a holistic solution for implementing cloud-enabled applications and service interactions in IoT scenarios in a fashion that improves the overarching goal of attaining end-user trust and, in turn, improve user adoption of IoT applications (Apps).

We consider some of the key spheres of significance in arriving at a reference architecture that is aimed at achieving trustworthiness among end-users in IoT applications, as being reminiscent of the implementation of security and privacy in:

- The *IoT application*, holistically

- Ubiquitous computing systems in the solution
- Participating *Cloud computing* systems
- In the *Service-Oriented Architecture* (SOA) layer

As the adoption race for the aforementioned innovative computing paradigms continue to mount, researchers have uncovered some of the critical concerns and proposed solutions to various facets of the all-inclusive problem that we seek to address. Among others, Itani et al. [1] described a set of security protocols for safeguarding privacy and compliance of end-user data in cloud computing scenarios. Yau et al. [2] looked at performance and security tradeoffs in SOA, while Ponemon Institute [3] shared an insightful research report detailing proactive steps to protect sensitive information in the cloud.

To throw some light on how various facets of security and privacy implementations can be considered in a holistic reference architecture for realizing end-user trust in IoT applications, we consider a case study where a cloud-hosted Movie Recommendation Service (achieved by leveraging both YouTube and Facebook Services) draws end-user media content viewing behavior and preferences in a multi-member family household from a TV-mounted Microsoft Kinect sensor device, a Smart TV and an Apple iPad tablet device collaboratively. The Microsoft Kinect Sensor is used to collect information about individual user preferences for movies in a multi-member family household. The movie recommendations service determines recommendations for movies that each household member or a combination of household members might enjoy based on the previous viewing history of each individual household member (captured by the Kinect Sensor and the iPad tablet device) in conjunction with the content viewing preferences of a given household member's influential Facebook friends. The ubiquitous Kinect Sensor device is used to identify specific users assembled in front of the family TV. Information gleaned by the Kinect sensor is transmitted through a mobile service to the cloud storage location. The Smart TV records the time of day that a movie was watched by a given household and saves it to the cloud storage through a cloud-hosted mobile service.

We describe our motivation for this study in section II and explain our key contributions through the:

- Explanation of some of the quintessential characteristics of a reference architecture for IoT Apps that seek to support privacy and security at all major layers of the IoT solution, in section III.

- Depiction of our proposed reference architecture for supporting privacy and security in IoT systems in section IV. We consider this design artifact as one of our key contributions.
- The implementation details for our case study are also described in section V.
- Comparison of our approach with that of previous research work is presented in section VI.

In section VII, we describe our evaluation and findings regarding the perception of trust among end-users in our IoT system prototype which seeks to implement our proposed reference architecture in the case study. In the light of our findings, we share our conclusions in section VIII.

II. MOTIVATION

This study advances our current investigations concerning the application of cloud services and collaborative ubiquitous devices in mobile health (mHealth) intervention scenarios. In most instances, sensitive health-related data is collected and stored in a cloud-hosted solution. The reference architecture presented in this paper provides a framework for ensuring that privacy and security take center-stage throughout the application development lifecycle in the pursuit of maximizing the promise of IoT, Ubiquitous computing and Cloud computing archetypes.

In addition, we seek to share a blueprint for developing software architecture that supports privacy and security in IoT solution designs. We envision that our template solution for the IoT domain can be adopted, refined and extended by software designers, engineers, and architects who seek to preserve trust across IoT implementations. By promoting the adoption of this reference architecture, we hope to institute a foundation for ensuring consistency in addressing privacy and security concerns among current and future IoT implementations. Previous studies [4] suggest that establishing a reference software architecture promotes re-use, reduces maintenance costs, and serves as a benchmark for software governance. We expect this reference architecture to serve IoT-related solution designers in the same capacity.

III. KEY CHARACTERISTICS OF THE CONCEPTUAL REFERENCE ARCHITECTURE

The reference software architecture for a given domain seeks to define the underlying components of the domain and their associated relationships [4]. The software architecture of a given implementation in the domain then becomes an instance of the domain reference architecture.

The *Internet of Things* can be described as an amalgamation of multiple heterogeneous digital devices, people, and services working collaboratively to solve a problem across technology boundaries, with the ability to seamlessly interact and share data about themselves and their environment [6]. In deriving our reference architecture, we outline the major components of modern IoT systems that can benefit from preserving security and data privacy by considering some of the fundamental components of IoT systems in the following applicable scenarios:

- *SCENARIO 1: A home automation monitoring service* that is capable of observing the usage of electricity in a given household and seamlessly regulates the home's consumption of electricity by learning the preferences of the household in comparison with optimal power consumption best practices of other neighboring households of similar size
- *SCENARIO 2: An Online Social Networking (OSN) service* that is integrated with a pervasive eyewear device, like Google Glass, for capturing pictures and recording videos of interesting moments. The service is assumed to be useful for storing the recorded photos and videos in cloud storage while enabling the end-user to share the stored media with other friends in the OSN as well as public access
- *SCENARIO 3: A movie recommendation service* that leverages previous media content viewing patterns and the preferences of influential people in a particular household user's circle of OSN friends to recommend future movies that will be of interest to the user. This scenario is employed in our case study implementation. A contextual view of the IoT-based movie recommendations scenario is illustrated below in Figure 1.

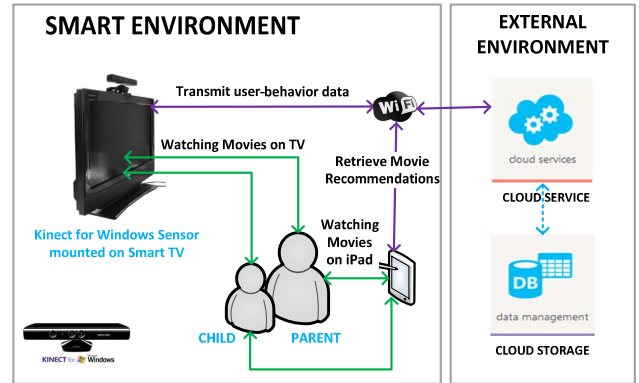


Fig. 1. Logical View of Interactions in the IoT system

In line with the scenario presented above, some of the major components that have their own facets for privacy and security concerns include:

- *End-User Preferences* for Security, Privacy and Trust
- *Cloud Computing*: in the form of a cloud-hosted web or mobile service and cloud-based data storage
- *Ubiquitous Computing*: represented by the Kinect Sensor, Tablet device and a Smart TV
- *Service Oriented Architecture (SOA)*: in the form of the Facebook Graph API (web service) used for inferring the preferences of influential friends in a given household member's OSN circle as well as the YouTube API (web service) for streaming a movie
- *Network communication* across wireless networks for transmitting and receiving data between the

ubiquitous devices in the Smart Environment and the external cloud service environment
Conceptually, most of the major facets of a generalized IoT implementation are likely to include the fundamental components captured in Figure 2 below.

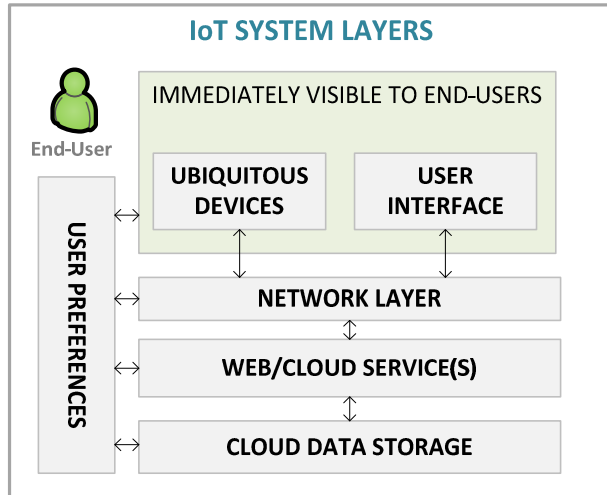


Fig. 2. Major facets for Security and Privacy in IoT App Scenarios

End-users have preferences for security, privacy and trust that must be collected and adhered to at all facets of the solution. There is typically an optional user interface and a physical sensor or ubiquitous device for data collection. These physical data communication components communicate to external systems through a network communication layer using a communication protocol to a web or mobile service of some nature. These web services, in turn, persists streams of data to a backend storage device. Typically, the backend service engine and storage is hosted in the cloud.

IV. REFERENCE ARCHITECTURE

In creating a generalized reference architecture for improving security and privacy concerns in IoT systems we illustrate the various layers and characteristics discussed in section III along with some of the best practices employed in security and privacy implementations in Figure 3.

In most cases, end-users are likely to accept an IoT solution that is managed or hosted on a trusted cloud provider system. We propose the use of a governance body for ongoing certification and regulation of standards pertaining to the all-encompassing extent of a typical IoT implementation.

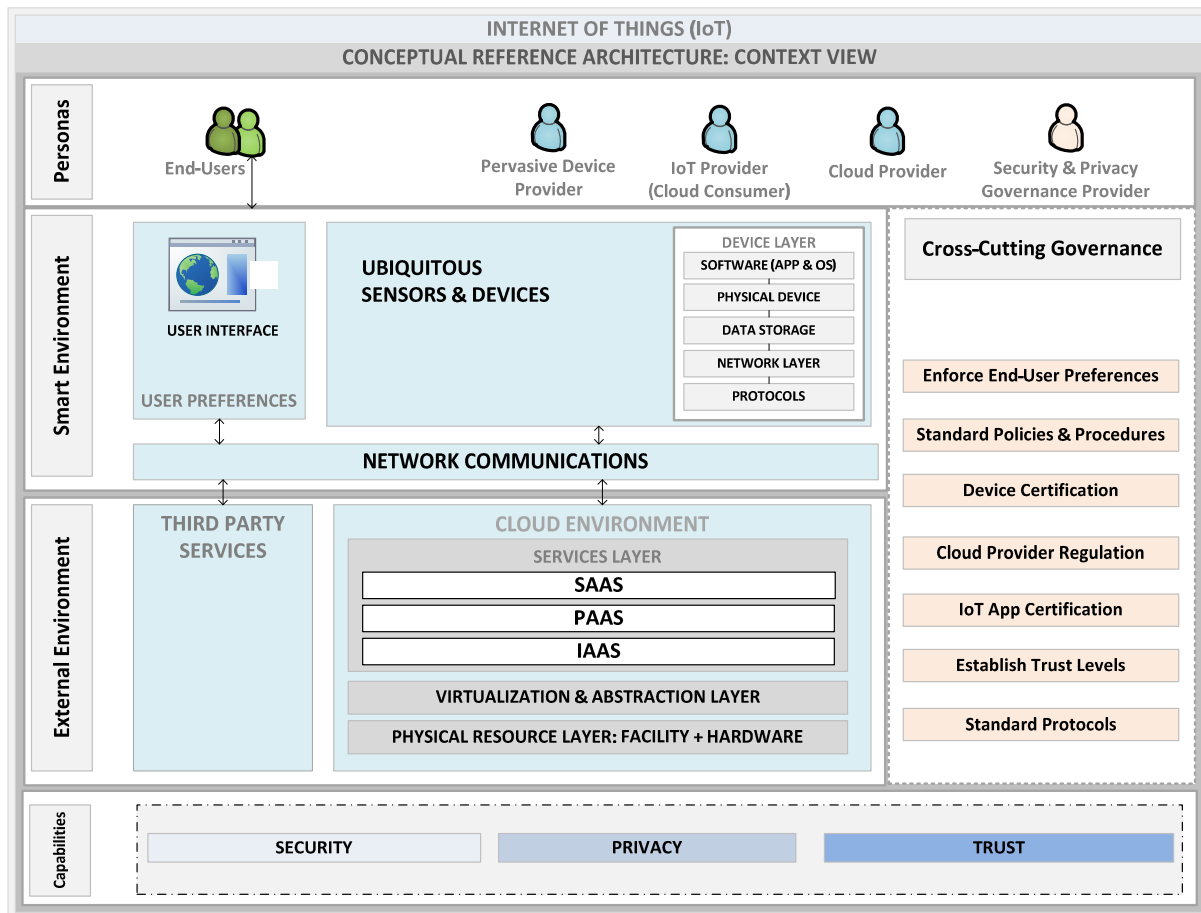


Fig. 3. Conceptual Reference Architecture: IoT

The ensuing literature describes some of the key concerns inherent in each layer of the IoT reference architecture.

A. Security and Privacy in the Ubiquitous Sensors and Devices in the Smart Environment

In considering the security and privacy concerns of IoT applications, it is important to hone in on some of the security and privacy challenges pertaining to pervasive devices and sensors that are often working ubiquitously to collect and exchange data in the environment. From a security and privacy perspective, some of the key requirements that can be addressed at this layer of the IoT application include [6]:

- *User identification and validation* to control access and enforce permissions and authorization levels for various components of the system
- *Tamper resistance* of the physical and logical device. Because IoT devices are typically unattended, physical attack vulnerabilities are critical.
- *Content security* - through digital rights management (DRM) of content used in the system
- *Data privacy* to protect sensitive user data.
- *Data communications and storage security* through protective measures for both data in-transit and data at-rest
- *Secure network communications* to ensure that network communications between ubiquitous devices and external services are only authorized through secure connection channels (for example, the wireless communication in the smart home environment of our case study can only be transmitted through the user's designated "home" wireless router, by default. Eavesdropping vulnerabilities in the wireless network must be curbed.
- *Privacy in ubiquitous computing* comes to play because the way in which the device or sensor collects data about the end-users might conflict with the user's privacy preferences for a particular scenario. For example, while a user might be open to having the Kinect sensor perform user identification to support the IoT system, he or she might not be open to having the Kinect record certain conversations in the smart home living room. These privacy barriers and preferences must be preserved in order to instill end-user trust in the system

That notwithstanding, some of the key security features that can be incorporated in pervasive device architecture (that is considering both hardware and software requirements) include [6]:

- Lightweight cryptography approaches to support the low power, memory and processing power constraints in most pervasive sensors and actuators
- Security measures to protect the physical device
- An interface for determining and controlling the privacy preferences of the end-user. This might include a visual cue (for example, a green light) when the device is in recording mode. The ability to stop or

pause the recording mode easily becomes critical as well

- Standard security protocols for direct device-to-device communication and device-to-service communication. For example, the Smart TV in our case study might need to send the user data securely to the cloud service in our case study through a secure socket layer (SSL) protocol over HTTPS communication.
- Secure on-device storage including RAM, Flash or ROM storage
- Secure operating system to protect data during runtime execution

In everyday human interactions, trust is often demonstrated when a user is able to confide in a close friend by sharing private or confidential information with the friend knowing that the friend will respect the display of trust by protecting the information from leaking to other unauthorized parties. Consequently, privacy goes hand-in-hand with trust. Privacy is sometimes defined as a critical human right "to be left alone" or an elementary need for an end-user to establish and maintain a private barrier to protect his or her information [7]. In devising a privacy solution at this level, Solove's [8] taxonomy for addressing privacy might be considered during information collection, information processing, information dissemination, and in curbing invasion.

In smart pervasive environments, the information collection process is often invisible to the end-user by virtue of the built-in ubiquity of the solution. This often raises concerns among user privacy experts. In most cases, the end-user does not have access to the information that has been collected about his or her activities. Equally, the end-user often has minimal visibility and control over the processing and dissemination of the data collected in the smart environment. However, IoT promotes a lot of data sharing among multiple devices. It is more critical for devices in an IoT setting to conform to standard privacy protocols that provide end-users with a level of comfort and trustworthiness that their private data will be protected. These issues have to be considered in deriving an effective architecture for hardware and software privacy.

B. Security and Privacy in the Cloud Computing Layer

Cloud computing can be defined as "a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction" [9]. In a typical cloud system model, there is a [1] Cloud provider – who exposes cloud services, and a Cloud consumer – who consumes these services. In our reference architecture, the IoT App Provider is also considered as a Cloud consumer.

Vulnerabilities in cloud systems can be categorized as [13] being related to Cloud Multi-tenancy, Elasticity, Availability of information (SLA), Information Integrity and Privacy, Secure Information Management, and Cloud Secure

Federation. It is important the IoT provider considers these vulnerabilities in arriving at a secure solution.

Nonetheless, vulnerabilities in cloud solutions can differ for a given cloud deployment model. Some of the cloud deployment models in use today include [14] [15]:

- Private Cloud
- Community Cloud
- Public Cloud
- Hybrid Cloud
- Virtual Private Cloud

Beyond the considerations for each cloud deployment model, there are unique security and privacy issues in each delivery model. We consider the following layers in an IoT solution that makes use of a *public cloud* deployment solution:

- *Services Layer* - which includes:
 - Software Applications
 - Data management systems
 - Operating Systems
- *Server Virtualization layer*
- *Physical Hardware layer* - which includes:
 - Physical hardware
 - Network communication infrastructure

Some of the popular cloud delivery models [13] [14] include:

- *Infrastructure-as-a-Service (IaaS)*: Where the cloud provider offers storage and computing services on-demand. The cloud customer manages the virtual machines (VMs) and other associated infrastructure components hosted in the cloud – including data storage, operating system and applications hosted on the VMs. Resultantly, security and privacy concerns at the application and operating system level are managed by the cloud customer. The cloud provider will handle security and privacy concerns at the datacenter hosting level including the virtualization and physical hardware layers.

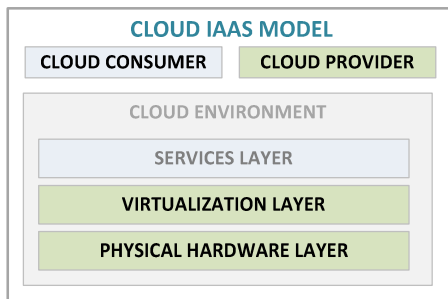


Fig. 4. Logical View of layers in Pervasive Devices

- *Platform-as-a-Service (PaaS)*: In this delivery model, the cloud provider exposes a set of services and application protocol interfaces (APIs) for developers to host web sites and services without having to deal with the scalability issues of the application as the solution usage grows. The cloud customer will have minimal control over the security practices used at the operating system-level.
- *Software-as-a-Service (SaaS)*: In this scenario, the cloud provider exposes specific applications to

consumers for use with a multi-tenancy approach that might use a subscription-based pay-per-use model. The Cloud Consumer will have even less control over the privacy and security implementation in this cloud-hosted solution since the entire service layer is managed by the cloud solution provider.

For a cloud consumer in an IoT setting, the trust model detailed in Itani and Kayssi's findings [1] can prove to be useful:

- *Full Trust* – where insensitive data is safely transmitted, stored, and processed without encryption on the cloud service
- *Compliance-based Trust* – where sensitive consumer data needs to be encrypted and sometimes anonymized in support of legal compliance regulations. An example might be compliance requirements for the Health Insurance Portability and Accountability Act (HIPAA) in mHealth scenarios.
- *No Trust* – where highly-sensitive customer data must be concealed from the cloud provider.

C. Security and Privacy in the IoT Apps and Service Layer

Security issues in integrating mobile agents and devices with services can be categorized as [18]: Confidentiality, Authentication, Authorization, Integrity, Nonrepudiation, Privacy, and Availability. In most typical IoT App scenarios multiple service endpoints were employed in the solution. The IoT system interacts with its own cloud-hosted service layer as well as external services. The IoT application user interface itself might have its own privacy and security concerns. In addition, the third party external services used in the solution might need to be governed to ensure that they protect the end-user's privacy and security preferences. For example, if the IoT application interacts with the Facebook Graph API, the end-user might have specific privacy settings set on Facebook (an OSN) that needs to be protected in the IoT system.

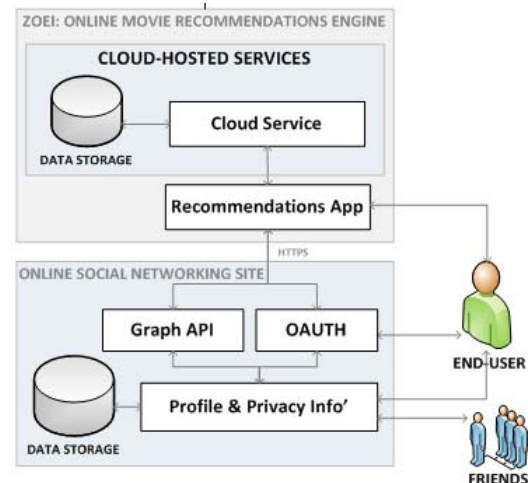


Fig. 5. External OSN service (Facebook) API interactions between an end-user and the IoT movie recommendations application

D. Cross-Cutting Governance Layer

In investigating an overarching solution that can protect end-users from the security and privacy vulnerabilities inherent in pervasive and cloud systems, we believe that it might be useful to introduce a trusted third-party security and privacy governance organization and/or a standard protocol for monitoring and developing ongoing compliance requirements that can be used to certify and regulate cloud and ubiquitous monitoring device providers in a bid to garner the kind of trust that is needed in such a big sphere of concerns.

From a cloud provider perspective, Ponemon Institute [3] recommends a number of proactive steps to be taken to protect sensitive information in the cloud environment including:

- Employ policies and procedures that stress the importance of protecting sensitive data in the cloud
- Assess the security status of third party vendors prior to sharing sensitive information through a thorough audit or review of the vendors' security qualifications
- Establish a functional role dedicated to information governance oversight to ensure better security practices are employed
- Provide transparency into the security infrastructure to help instill confidence in cloud consumers that information stored in the cloud is secure

We believe that a third-party governance institution that can address the need for third-party assessment, regulations at all layers of IoT systems, transparency, policies, standards and ongoing status certification will be greatly beneficial to improving the trustworthiness of IoT applications.

V. CASE STUDY: MOVIE SUGGESTIONS IoT APP

We implemented a prototype solution for a personalized movie recommendation IoT Application that takes into account our proposed reference architecture in order to derive the IoT Application's solution architecture for governing security and privacy concerns at various layers of the solution.

Personas or Actors identified in the reference model include:

- End-users of the IoT solution
- Device providers or vendors who build pervasive sensors and devices that are used in IoT systems. For the Kinect sensor, the device vendor is Microsoft.
- IoT Application Provider and Cloud Consumer
- Cloud Provider who exposes cloud services that are used by IoT Providers. For our case study, the Cloud Provider was Microsoft's Azure Cloud. The third-party OSN provider in our case study was Facebook's Graph API while movie trailers surfaced in the App were served through Google's YouTube API.
- Our pseudo third-party Security and Privacy governance provider is expected to take on the regulation and certification of various components in the IoT context including the cloud service and storage, external services like the Facebook API, and

the IoT application itself. The provider also ensures adherence to standard protocols like OAUTH, SSL, etc. across the communication channels.

The external services employed include the Facebook Graph API and the YouTube API. To access the household member's Facebook friend information the privacy of the user's data must be considered in the external system. Access to the end-user's Facebook information is controlled by Facebook's implementation of a secure OAUTH 2.0 over HTTPS protocol for user authentication. In addition, Facebook allows the household user to authorize specific access to the IoT system, after the user's credentials have been successfully verified. An access token is required by both the YouTube API and the Facebook API for the IoT endpoint to exchange data with the third party services.

For our case study, the IoT application's user interface takes the form of a movie recommendations engine accessible through both an iPad device and a Smart TV as a web-based application. The user interface for our movie recommendations IoT App is represented in Figure 6 below.



Fig. 6. Screenshot of the Movie suggestions App

The network communications component in the smart environment, in our case study, represents a wireless router for secure communications between the smart devices and cloud-hosted IoT web service. The ubiquitous computing device in our case study is represented by the Kinect for Windows sensor connected to the Smart TV.

The movie suggestions that are provided in the application are ranked based on the social influence of a particular household user's OSN friends as well as the previous movie preferences of the OSN friends (captured as Facebook "Likes"). In addition, movies that are similar to media content

that was previously identified by the Kinect sensor as part of the authenticated user's viewing history are factored into ranking the recommended movies for that particular user. While the benefit of this scenario seems to outweigh the need for the Kinect sensor to monitor the user's viewing behavior, there can be a number of privacy and security concerns that might deter end-user adoption.

The cloud system model used in our case study analysis is synonymous with that of a *public cloud* [1] where the cloud provider manages and exposes storage and computing services through a geographically dispersed and on-demand scalable cloud infrastructure. We (as the cloud consumer) then utilize cloud storage to host data gleaned from the IoT device sensors. The web services employed in the IoT scenario also operate as cloud services hosted on the *PaaS cloud* infrastructure.

Improving Privacy and Security Concerns in our Case Study:

Some of the concerns considered in enforcing privacy and security best practices at various layers of the reference architecture include:

- **Informed Consent:** We learned that end-users preferred to be notified when the Kinect sensor is collecting both sensitive and non-sensitive data in the smart environment. A visual cue by the form of a blinking green light indicator on the device while it is in recording mode proves to be useful.
- **Control over Privacy Settings:** The parent's in the household may not want their children to have access to uncensored content, so the parent might want control over media content suggestions that are surfaced to the children in the household. Also, the parent might want to limit how much data is stored by the IoT App (for example, exclude geo-location information from data collection).
- **Vendor Regulation:** A third-party regulation body could be employed to monitor and expose gaps in the system based on the end-user's pre-defined security and privacy preferences. Ongoing risk assessments on behalf of the end-users could prove to be useful.
- **Access to User Data and Opt-Out:** In some cases, participants prefer control over the data that is collected about them. End-users are also interested in how their data is used and seek to reserve the ability to opt-out and delete their data at will.
- **Ongoing Reputation Access:** Beyond the participant's initial consent to allowing the IoT App access to his or her Facebook (OSN) data, it will be useful if the participant can access the privacy of the IoT solution itself at any point in time and opt-out without any loom of lock-in.
- **User Identification and Authentication:** If the identity management system throughout the IoT implementation is not accurate, uncensored content that might be appropriate for the parent but inappropriate for the child might be surfaced mistakenly to the child.

- **Physical Security and Wireless Networks:** Prevention of eavesdropping in the wireless network as well as measures to enforce security of the physical objects in the environment proves to be critical.

VI. RELATED WORKS

While several security and privacy frameworks and reference architectures exist for various domains, we are not aware of a generic reference architecture that caters to modern IoT App scenarios. NIST [22] proposed a reference architecture for Cloud Computing and Itani et al. [1] have explored a reference framework for privacy in cloud computing scenarios. Several studies have focused on legal compliance and the technical implementation of trust models, frameworks and protocols. Additionally, Langheinrich [23] addressed key privacy concerns in ubiquitous systems.

TABLE I. COMPARISON WITH OTHER APPROACHES

CHARACTERISTICS	NIST [22]	Langheinrich [23]	Our proposed reference architecture
Ubiquitous sensor and device concerns	No	Yes	Yes
Cloud Computing concerns	Yes	No	Yes
IoT domain concerns	No	No	Yes
Includes both Security and Privacy concerns	Yes	No - privacy only	Yes
Proposes a third-party governance	Yes – cloud auditor	No	Yes

VII. EVALUATION

To test the hypothesis that our reference architecture is feasible for most modern IoT implementations, we reviewed the implementation of security and privacy solutions aimed at achieving trust in the three scenarios (among others) described in section III and compared the inherent layers in the scenario to that of our proposed conceptual reference architecture.

In addition, we built a prototype of the case study described in this paper and shared the way in which various security and privacy concerns are addressed in our work-in-progress solution with a few technologically savvy end-users. We collected feedback from these end-users through a survey to gauge their comfort level with the implementation of security and privacy solutions at various layers of the conceptual model.

Of the 18 distributed surveys, we received 14 usable responses with questions centered on evaluating the importance of the various facets in the reference architecture and how it affects their overall level of trust in the IoT system:

- *Trust in the underlying Cloud data storage system*
- *Trust in the ubiquitous devices and user interface*

- Trust in the vendor services involved in the IoT App
- Need for a third-party regulating body

Participants (end users) were asked to indicate their rating with a scale of 1 through 3 (where 3 means the characteristic is *Important*, 2 represents *Indifferent*, and 1 denotes *Not Important*). Our analysis of the survey response indicates that most savvy respondents (92.8%) were most comfortable with the IoT system when the immediately visible device or sensor exhibited security and privacy best practices.

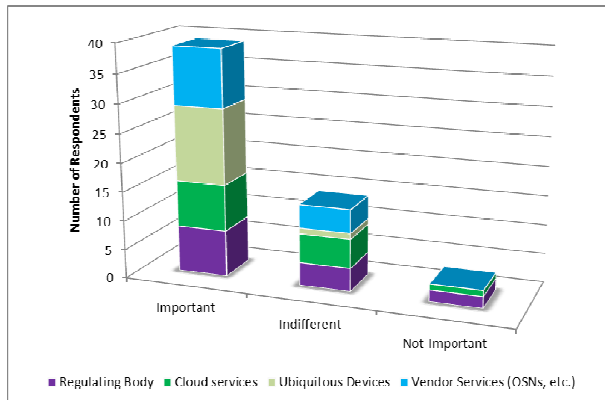


Fig. 7. Savvy End-User Response

VIII. CONCLUSIONS

Based on the findings of our user study, even though addressing key security and privacy concerns holistically helps to minimize end-user adoption barriers, perceptions related to the trustworthiness of an IoT application hangs significantly on the implementation of security and privacy best practices in the immediately visible IoT device or application user interface. Nevertheless, with growing concerns of security and privacy in IoT Application scenarios, we believe that the proposed reference architecture can be adopted by researchers and IoT solution architects, at large, as a yardstick for governing the implementation of security and privacy concerns at all facets of an overarching IoT solutions architecture.

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